



Qualification Pack



QUALIFICATION FILE

Green Energy Supervisor – Solar

☒ Short Term Training (STT) ☐ Long Term Training (LTT) ☐ Apprenticeship

☒ Upskilling ☐ Dual/Flexi Qualification ☐ For ToT ☐ For ToA

☒ General ☐ Multi-skill (MS) ☐ Cross Sectoral (CS) ☐ Future Skills ☐ OEM

NCrF/NSQF Level: 5

Submitted By:

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GREEN ENERGY SUPERVISOR - SOLAR

QP Code: TPCL/SGJ/Q0301

Version: 1.0

NSQF Level: 5

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TPCL/SGJ/Q0301: GREEN ENERGY SUPERVISOR - SOLAR

Brief Job Description

A Green Energy Supervisor Solar specializes in technology, installation, and maintenance & troubleshooting of photovoltaic (PV) systems to ensure efficient solar energy generation. They are responsible for implementing system design, installing panels, troubleshooting issues, and performing routine maintenance to enhance performance and safety. Their role also involves adhering to industry standards, optimizing energy output, and contributing to the growth of renewable energy solutions.

Personal Attributes

Leadership, Team Management, Attention to Safety & Compliance, Problem-Solving Mindset and Technical Aptitude

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TPCL/SGJ/N0303: Understand and implement Safety and Health Practices in Solar PV Projects](#)
2. [TPCL/SGJ/N0301: Interpret and Apply Electrical Concepts and Wiring](#)
3. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)
4. [TPCL/SGJ/N0302: Demonstrate knowledge of Renewable Energy and Execute the Solar PV Plant Lifecycle](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	GREEN ENERGY SUPERVISOR – SOLAR
Country	India
NSQF Level	5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NSQF

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Minimum Educational Qualification & Experience	12th grade pass with 2 year NTC/ CITS/NAC with 3-5 Years of experience OR Pursuing 1st year of UG and continuous education OR Pursuing 2nd year of 3-year regular Diploma (after 10th) with NA of experience OR Previous relevant Qualification of NSQF Level (4.5) with 1-2 Years of experience OR Previous relevant Qualification of NSQF Level (4) with 2-3 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	17 Years
Last Reviewed On	NA
Next Review Date	07/05/2028
NSQC Approval Date	08/05/2025
Version	1.0
Reference code on NQR	QG-05-ES-03868-2025-V1-TATAPWR
NQR Version	1

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TPCL/SGJ/N0303: Understand and implement Safety and Health Practices in Solar PV Projects

Description

Covers safety practices including electrical safety, working at heights, material handling, fire safety, first aid, and housekeeping.

Scope

The scope covers the following :

- Applies to supervising safety and health practices during solar PV installation, O&M, and project execution.

Elements and Performance Criteria

General Safety

To be competent, the user/individual on the job must be able to:

PC1. Explain general workplace safety guidelines and best practices.

Electrical Safety

To be competent, the user/individual on the job must be able to:

PC2. Describe the hazards associated with electrical work.

Safety While Working at Height

To be competent, the user/individual on the job must be able to:

PC3. Explain the risks and precautions while working at heights.

Material Handling Safety

To be competent, the user/individual on the job must be able to:

PC4. Describe the proper techniques for lifting and transporting materials.

Behavioural Safety

To be competent, the user/individual on the job must be able to:

PC5. Explain the importance of a proactive safety culture in the workplace.

Fire Safety

To be competent, the user/individual on the job must be able to:

PC6. Explain fire hazards, fire classifications, and firefighting techniques.

First Aid

To be competent, the user/individual on the job must be able to:

PC7. Explain the basic principles of first aid and emergency response.

Housekeeping and 5S

To be competent, the user/individual on the job must be able to:

PC8. Explain the principles of 5S methodology and its benefits.

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** Knowledge of safety standards, PPE, LOTO, firefighting, first aid, safe material handling, and emergency response.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to enforce safety compliance, manage risks, and promote safe working culture on solar PV sites.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>General Safety</i>	1	2	-	-
PC1. Explain general workplace safety guidelines and best practices.	1	2	-	-
<i>Electrical Safety</i>	1	2	-	-
PC2. Describe the hazards associated with electrical work.	1	2	-	-
<i>Safety While Working at Height</i>	1	2	-	-
PC3. Explain the risks and precautions while working at heights.	1	2	-	-
<i>Material Handling Safety</i>	1	2	-	-
PC4. Describe the proper techniques for lifting and transporting materials.	1	2	-	-
<i>Behavioural Safety</i>	1	1	-	-
PC5. Explain the importance of a proactive safety culture in the workplace.	1	1	-	-
<i>Fire Safety</i>	1	2	-	-
PC6. Explain fire hazards, fire classifications, and firefighting techniques.	1	2	-	-
<i>First Aid</i>	1	2	-	-
PC7. Explain the basic principles of first aid and emergency response.	1	2	-	-
<i>Housekeeping and 5S</i>	1	1	-	-
PC8. Explain the principles of 5S methodology and its benefits.	1	1	-	-
NOS Total	8	14	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	TPCL/SGJ/N0303
NOS Name	Understand and implement Safety and Health Practices in Solar PV Projects
Sector	Green Jobs
Sub-Sector	
Occupation	GREEN ENERGY SUPERVISOR - SOLAR
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TPCL/SGJ/N0301: Interpret and Apply Electrical Concepts and Wiring

Description

Covers fundamentals of electricity, circuit diagrams, wiring practices, and application of electrical concepts in solar PV systems.

Scope

The scope covers the following :

- Applies to supervising and implementing basic and advanced electrical wiring for solar PV installation and maintenance.

Elements and Performance Criteria

Fundamentals of Electricity

To be competent, the user/individual on the job must be able to:

- PC1.** Identify basic electrical terms: voltage, current, and resistance.
- PC2.** Explain the use of Ohm's Law in simple circuits.
- PC3.** Demonstrate the measurement of voltage, current, and resistance using a multimeter.

Circuit Diagrams and Wiring

To be competent, the user/individual on the job must be able to:

- PC4.** Identify common electrical symbols in circuit diagrams.
- PC5.** Explain the difference between series and parallel circuits.
- PC6.** Demonstrate basic electrical wiring for a simple circuit.
- PC7.** Demonstrate how to check continuity in a circuit using a multimeter.
- PC8.** Identify safety precautions for electrical wiring.
- PC9.** Demonstrate proper wire connections and insulation techniques.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand electrical concepts, conductors, insulation, earthing, fuses, circuit diagrams, electromagnetism.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to perform wiring, use measuring instruments, ensure safety compliance, and supervise electrical work.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fundamentals of Electricity</i>	2	2	-	-
PC1. Identify basic electrical terms: voltage, current, and resistance.	1	-	-	-
PC2. Explain the use of Ohm's Law in simple circuits.	1	-	-	-
PC3. Demonstrate the measurement of voltage, current, and resistance using a multimeter.	-	2	-	-
<i>Circuit Diagrams and Wiring</i>	6	10	-	-
PC4. Identify common electrical symbols in circuit diagrams.	2	-	-	-
PC5. Explain the difference between series and parallel circuits.	1	1	-	-
PC6. Demonstrate basic electrical wiring for a simple circuit.	-	3	-	-
PC7. Demonstrate how to check continuity in a circuit using a multimeter.	-	3	-	-
PC8. Identify safety precautions for electrical wiring.	2	-	-	-
PC9. Demonstrate proper wire connections and insulation techniques.	1	3	-	-
NOS Total	8	12	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	TPCL/SGJ/N0301
NOS Name	Interpret and Apply Electrical Concepts and Wiring
Sector	Green Jobs
Sub-Sector	
Occupation	GREEN ENERGY SUPERVISOR – SOLAR
NSQF Level	5
Credits	5
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

TPCL/SGJ/N0302: Demonstrate knowledge of Renewable Energy and Execute the Solar PV Plant Lifecycle

Description

Focuses on renewable energy basics, solar PV design, installation, commissioning, O&M, troubleshooting, and performance monitoring.

Scope

The scope covers the following :

- Applies to supervising and executing end-to-end solar PV project lifecycle from site survey to O&M.

Elements and Performance Criteria

Fundamental Concepts of Renewable Energy

To be competent, the user/individual on the job must be able to:

PC1. Explain the different types of renewable energy sources and their significance.

PC2. Describe the advantages and limitations of renewable energy sources.

Fundamentals of Solar Photovoltaic

To be competent, the user/individual on the job must be able to:

PC3. Explain the basic working principle of solar photovoltaic technology.

PC4. Describe the different types of solar panels and their efficiency factors.

Tools and Components for Solar PV Systems

To be competent, the user/individual on the job must be able to:

PC5. Identify the essential tools required for solar PV installation.

PC6. Explain the function of key electrical and mechanical components in a solar PV system.

Site Survey and Assessment

To be competent, the user/individual on the job must be able to:

PC7. Explain the importance of conducting a site survey before solar PV installation.

PC8. Demonstrate the use of site survey tools and techniques for solar PV assessment.

Installation and Commissioning of Solar PV Sys

To be competent, the user/individual on the job must be able to:

PC9. Explain the step-by-step process of installing a solar PV system.

PC10. Describe the commissioning process and performance verification of a solar PV system.

Operation and Maintenance of Solar PV Systems

To be competent, the user/individual on the job must be able to:

PC11. Explain the key maintenance practices for solar PV systems.

PC12. Describe fault detection techniques for solar PV systems.

Solar PV System Performance Monitoring & Troubleshooting

To be competent, the user/individual on the job must be able to:

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PC13. Explain the importance of monitoring solar PV system performance.

PC14. Demonstrate basic troubleshooting of solar PV systems.

Solar PV Financials, Procedures & Permissions

To be competent, the user/individual on the job must be able to:

PC15. Explain the key financial aspects of solar PV projects, including ROI and payback period.

PC16. Describe the regulatory approvals and procedures for solar PV installations.

New Trends in Solar PV

To be competent, the user/individual on the job must be able to:

PC17. Explain emerging technologies in solar PV, such as bifacial panels and floating solar.

PC18. Describe advancements in energy storage and hybrid solar solutions.

Overview of Manufacturing

To be competent, the user/individual on the job must be able to:

PC19. Explain the basic process of solar PV panel manufacturing.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand renewable energy concepts, solar PV system components, installation procedures, monitoring methods, and financials.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to supervise installation, optimize system performance, troubleshoot issues, and manage solar PV lifecycle effectively.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fundamental Concepts of Renewable Energy</i>	4	4	-	-
PC1. Explain the different types of renewable energy sources and their significance.	2	2	-	-
PC2. Describe the advantages and limitations of renewable energy sources.	2	2	-	-
<i>Fundamentals of Solar Photovoltaic</i>	4	4	-	-
PC3. Explain the basic working principle of solar photovoltaic technology.	2	2	-	-
PC4. Describe the different types of solar panels and their efficiency factors.	2	2	-	-
<i>Tools and Components for Solar PV Systems</i>	4	4	-	-
PC5. Identify the essential tools required for solar PV installation.	2	2	-	-
PC6. Explain the function of key electrical and mechanical components in a solar PV system.	2	2	-	-
<i>Site Survey and Assessment</i>	3	3	-	-
PC7. Explain the importance of conducting a site survey before solar PV installation.	2	2	-	-
PC8. Demonstrate the use of site survey tools and techniques for solar PV assessment.	1	1	-	-
<i>Installation and Commissioning of Solar PV Sys</i>	2	2	-	-
PC9. Explain the step-by-step process of installing a solar PV system.	1	1	-	-
PC10. Describe the commissioning process and performance verification of a solar PV system.	1	1	-	-
<i>Operation and Maintenance of Solar PV Systems</i>	2	2	-	-
PC11. Explain the key maintenance practices for solar PV systems.	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Describe fault detection techniques for solar PV systems.	1	1	-	-
<i>Solar PV System Performance Monitoring & Troubleshooting</i>	2	2	-	-
PC13. Explain the importance of monitoring solar PV system performance.	1	1	-	-
PC14. Demonstrate basic troubleshooting of solar PV systems.	1	1	-	-
<i>Solar PV Financials, Procedures & Permissions</i>	1	2	-	-
PC15. Explain the key financial aspects of solar PV projects, including ROI and payback period.	1	1	-	-
PC16. Describe the regulatory approvals and procedures for solar PV installations.	-	1	-	-
<i>New Trends in Solar PV</i>	-	2	-	-
PC17. Explain emerging technologies in solar PV, such as bifacial panels and floating solar.	-	1	-	-
PC18. Describe advancements in energy storage and hybrid solar solutions.	-	1	-	-
<i>Overview of Manufacturing</i>	-	1	-	-
PC19. Explain the basic process of solar PV panel manufacturing.	-	1	-	-
NOS Total	22	26	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	TPCL/SGJ/N0302
NOS Name	Demonstrate knowledge of Renewable Energy and Execute the Solar PV Plant Lifecycle
Sector	Green Jobs
Sub-Sector	
Occupation	GREEN ENERGY SUPERVISOR – SOLAR
NSQF Level	5
Credits	7
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

pre test, post test and behavioral assessment

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TPCL/SGJ/N0303.Understand and implement Safety and Health Practices in Solar PV Projects	8	14	0	0	22	22
TPCL/SGJ/N0301.Interpret and Apply Electrical Concepts and Wiring	8	12	0	0	20	20
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
TPCL/SGJ/N0302.Demonstrate knowledge of Renewable Energy and Execute the Solar PV Plant Lifecycle	22	26	-	-	48	48
Total	58	82	-	-	140	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.